

Physics-Informed ML and Domain Adaptation for Dark Matter Classification in Gravitational Lenses

Pallab Mondal | Student ID: 946804 | AI Models for Physics 2025/2026 | May 2026

Summary

Strong gravitational lensing probes dark matter substructure via arc distortions. We compare ResNet-18, LensPINN (physics-informed CNN with differentiable SIS lens inversion), and HEALSwIn on DeepLense Model I (30,000 train / 7,500 test; 64x64; classes: no_substructure, cdm_subhalos, axion). Phase 2 applies ADDA domain adaptation and classifies real SDSS cutouts. Code: <https://github.com/Pallab9999/DeepLenseProject>

Method

ResNet-18 baseline: spatial convolutions on raw pixels. LensPINN: predict Einstein radius θ_E , invert $\beta = \theta - \alpha(\theta)$ with SIS $\alpha(\theta) = \theta_E * \theta / |\theta|$, reconstruct source I_{source} , form residual $I_{\text{residual}} = I_{\text{orig}} - I_{\text{source}}$; decoder uses original + source + residual. HEALSwIn: HEALPix + Swin attention + physics head. ADDA: freeze source encoder M_s , train target encoder M_t adversarially vs discriminator D to align simulated and telescope features.

Results (Phase 1: Model I)

Metric	ResNet-18	LensPINN	HEALSwIn
Accuracy	43.41%	52.35%	33.33%
Macro F1	0.355	0.511	0.167
CDM recall	2%	28%	0%
Smooth / Axion AUC	0.64 / 0.68	0.79 / 0.76	N/A

LensPINN improves accuracy by +9.0% absolute by subtracting the smooth Einstein ring and exposing subhalo shears. ResNet collapses on CDM (2% recall). HEALSwIn at 5 epochs shows mode collapse (33.3%): physics-informed models must learn θ_E before classification converges.

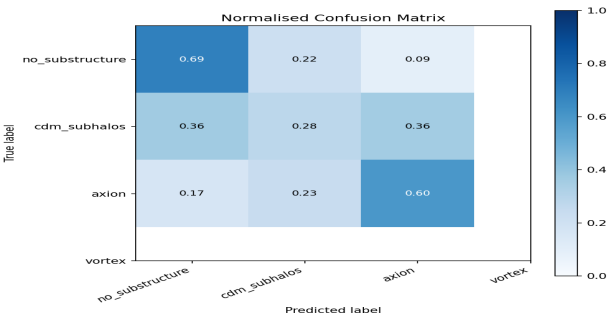


Fig 1: LensPINN confusion matrix

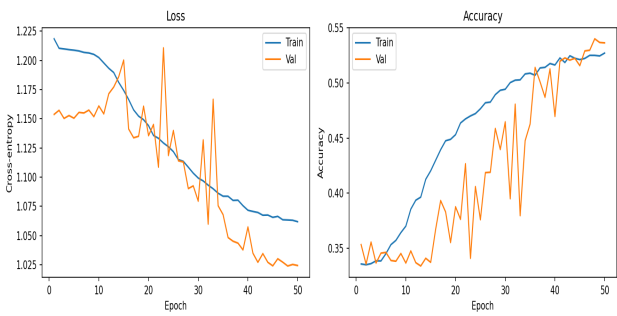


Fig 2: Training curves

Results (Phase 2: SDSS + ADDA)

Real RGB cutouts from SDSS SkyServer were grayscale-converted, center-cropped, and rescaled to 64x64. ADDA aligned target features to simulations. LensPINN classified Cosmic Horseshoe, Cheshire Cat, and Five-Image Quasar as no_substructure with >99.9% confidence - consistent with smooth macro-lenses at survey resolution (zero false-positive substructure detections).

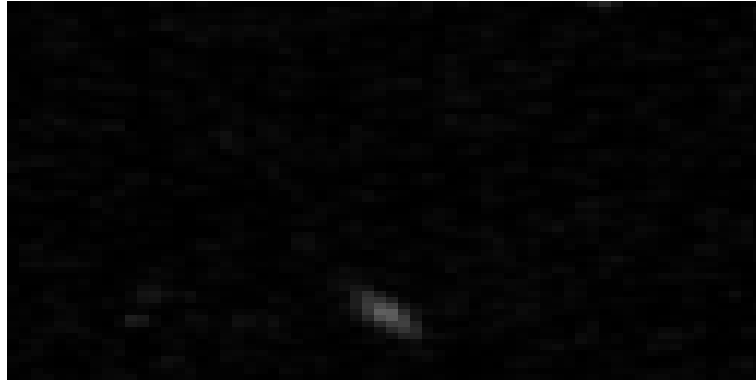


Fig 3: Preprocessed Cosmic Horseshoe (64x64); Cheshire Cat and Five-Image Quasar processed identically.

Conclusions

(1) Embedding SIS ray-tracing in the forward pass beats a pure CNN on subhalo detection. (2) ADDA plus coordinate-based SDSS ingestion enables sim-to-real inference. (3) Extended training is required for transformer-based physics models. Future work: JWST/LSST resolution and multi-class domain adaptation.

Key files for oral code discussion

<code>my_work/utils/physics.py</code>	- SIS deflection and inversion
<code>my_work/models/lens_pinn.py</code>	- theta_E head + residual channels
<code>my_work/train.py / evaluate.py</code>	- training and metrics
<code>my_work/train_adda.py</code>	- domain adaptation demo
<code>my_work/classify_real_lenses.py</code>	- SDSS inference